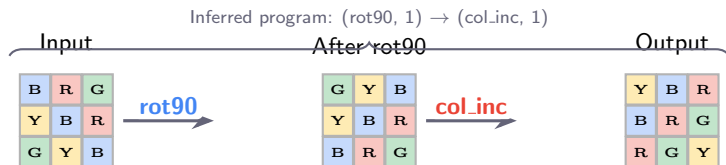


*ProgramEncoder infers composition from $K=3$
support pairs (input $\rightarrow$ output examples)*



Algebraic composition in HDC space:

$$\underbrace{z_{\text{out}}}_{\text{output}} = \underbrace{\text{col_inc}}_{\text{colour LUT}} \circ \underbrace{\text{rot90}}_{\text{cell permutation}} (z_{\text{in}}) \equiv P_{\text{rot}} \cdot z_{\text{in}} \text{ then LUT}[\text{colour} + 1]$$

Out-of-distribution generalisation (zero-shot, no retraining):

Novel rule	Composition	Accuracy
rotate_270	rot90 ³	100%
recolour_dec	col_inc ⁹	100%
flip_vertical	rot90 ² $\circ$ reflect	100%
shift_down	rot90 $\circ$ translate $\circ$ rot90 ³	100%